lens when you zoom in. The catch is that the underlying sensor beneath the telephoto lens is not the same sensor as the one on the primary camera. It's often a smaller sensor.

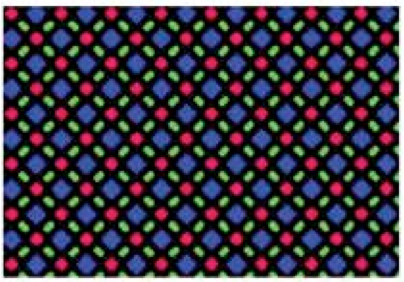
When you put a large focal length lens on a smaller sensor, the image appears zoomed in because the

smaller sensor uses less of the lens. The same 26mm lens on the primary camera can produce a zoom similar to a 39mm lens on a sensor half the size (1.5x crop factor).

So basically, what you get with a 2X zoom camera is equivalent to 2x zoom. The focal length of the smaller telephoto camera isn't actually 2x the focal length of the primary sensor. Also, the optical zoom available works only at one specific focal length. For instance, if your phone has 3x optical zoom, the zoom between 1x to 3x (1.5x, 2x, 2.5x, etc.) will all be digital zoom. This optical trick lets manufacturers market higher optical zoom by reducing the sensor size of the telephoto camera at the expense of quality. In fact, our phones use this trick on all sensors. If the primary camera advertises a 26mm lens, then it probably includes a 4.5mm lens on top of a tiny sensor and is actually 26mm equivalent.

YOUR DISPLAY MIGHT NOT BE AS SHARP

The actual resolution or sharpness of your AMOLED display might not be what is advertised. Almost all AMOLED displays on phones use a pentile pixel arrangement. What this means is that these displays have double green pixels than blue or red pixels and thus the effective resolution (Red, Green, Blue sub-pixel combos) is lesser than advertised. The red, green and blue subpixels are shared between pixels. No pixel has all three and every alternate pixel is missing a red or a blue. This is also why 2K AMOLED displays on Samsung phones in the past weren't significantly sharper than 1080P AMOLED or IPS LCD displays. Not just Samsung and other Android phones but also Apple iPhones use Pentile OLED displays. The pentile arrangement is popular because it helps increase the lifespan of the OLEDs and also helps OEMs advertise high resolution. Sure, there are artefacts because pixels are arranged in a criss-cross manner, but



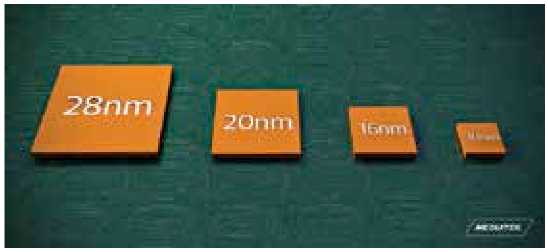
Source: Display Mate

Diamond Pentile sub-pixel layout of iPhone 13 Pro

their impact is drastically reduced at high resolutions. However, We do have a few RGB OLED displays on mobile devices like the one used on the Nintendo Switch OLED. Even Apple Watch models have a true RGB pixel matrix because when you need to represent physical objects like the moving hands of an analogue watch on a digital watch display, artefacts like aliasing are not pardonable.

THE NANOMETER IN A PROCESS NODE DOESN'T REFER TO ANYTHING

Every year flagship SoCs keep switching to smaller process nodes or technology nodes denoted by a nanometer number such as 7nm, 5nm, 3nm, etc. This number, however, lost its relevance several years ago



and doesn't mean anything as of today. There is no dimension that actually measures 7 nanometers on a chipset that is based on a 7nm process.

In the past, this number used to refer to the actual and effective gate length of the transistors in the chip and later to the M1 half pitch (or half the distance between metal 1 lines on a chip), but as of today it has lost its meaning and is more of a marketing term. The smaller the technology node, the smaller and more densely packed the transistors, which often results in better power efficiency. Just don't expect your transistors to shrink proportionally to the nanometer marker.

VRR ON PHONES IS NOT AS SEAMLESS AS IT SEEMS

This doesn't pose much practical inconvenience, but yes, VRR or variable refresh rate on phones is not as seamless as on monitors, TVs and PCs. Depending on the content being rendered, the displays on our phones essentially switch between a set of discrete available refresh rate options like 48 Hz, 60Hz, 90Hz and 120Hz. For instance, most phones can run at 48Hz or 60Hz but not at 53Hz or any in-between refresh rate.

Phones with faster LTPO displays have subsets within the set and thus can quickly switch between more levels. This helps conserve battery as the phone can switch to lower frame rates while displaying static content. With time as new display driver ICs evolve, we expect all phones to support seamless switching to any possible refresh rate between, say, 1Hz and 120Hz thus improving battery mileage. d